

Lab Assignment 2

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1 FE-Model

The geometry of the real beam was measured to be two 35x35 mm beams welded together forming a T-bone. The horizontal part was measured to be 327 mm from the fixed end to where the weld, and the vertical beam to be 500 mm. The two beams were hollow, with a thickness of 2 mm.

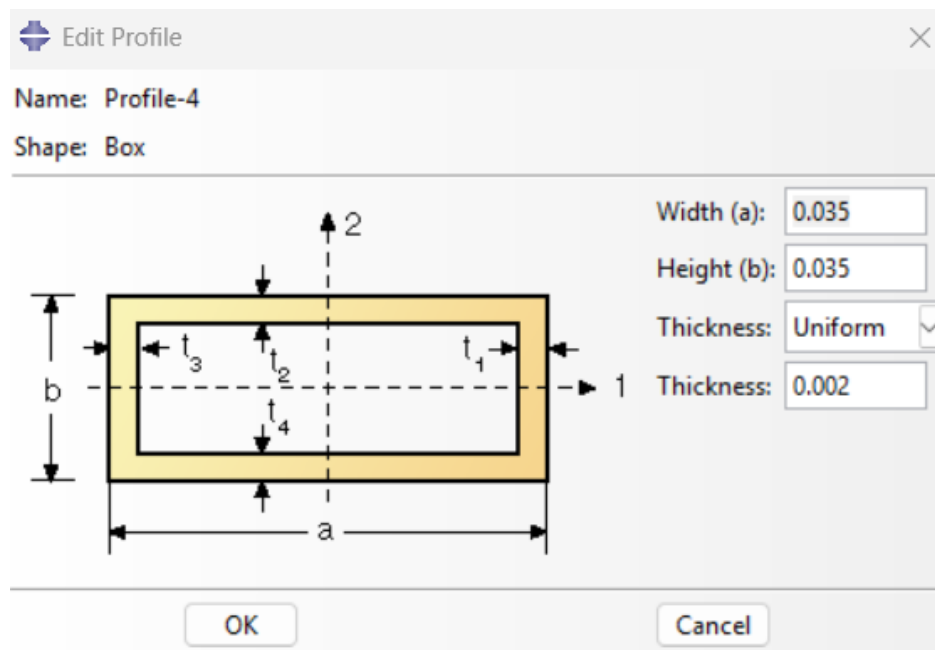


Figure 1: The dimensions set in Abaqus.

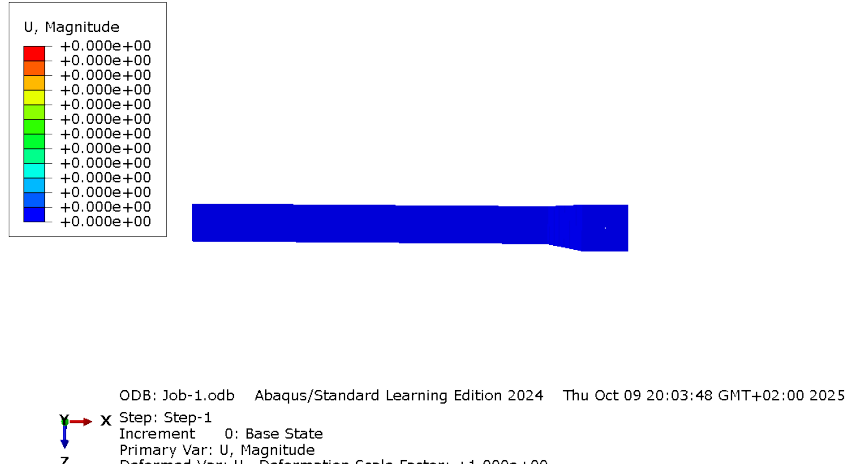


Figure 2: Base state

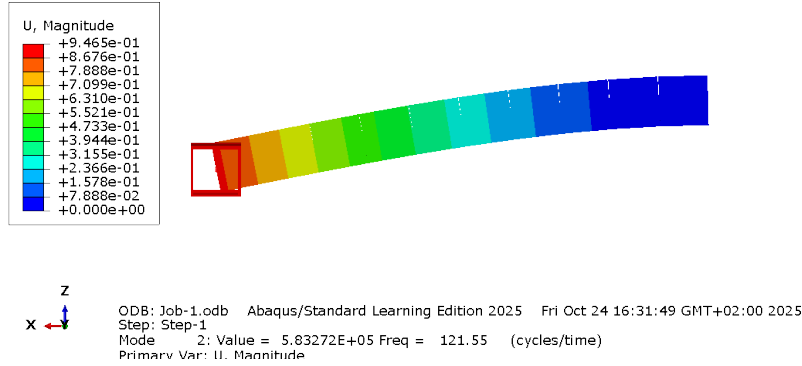


Figure 3: Mode shape 1

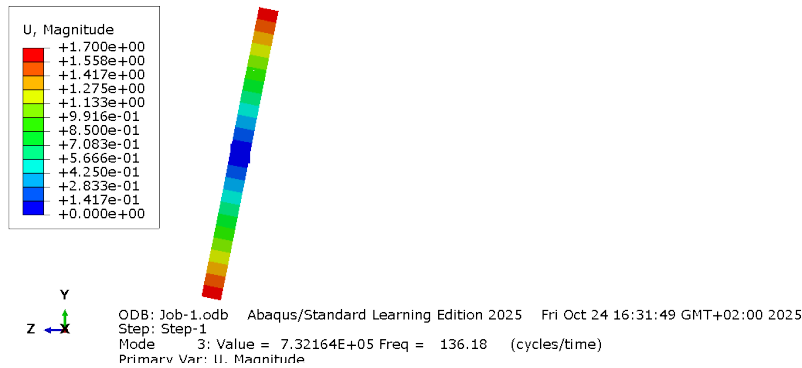


Figure 4: Mode shape 2

From [fig.3] and [fig.4], we can see that the first mode is most visible at any point on the vertical beam, while the second mode is most visible at the ends of the vertical beam. None of the modes are very visible close to the fixed end. Therefore the best place to place the shaker is at one of the ends of the vertical beam.

The FE model is a 3D deformable beam model. The FEM model in Abaqus gave the following natural frequencies:

Mode	Abaqus f [Hz]
1	121.55
2	136.18

Table 1: Resonant frequencies for first two modes - Abaqus

2 Measurement model

For the lab measurement we have chosen to place our excitation point at the bottom of the vertical beam, which is at point 1 in the figure below. The main reason is that we can easily excite the structure in that point, because the structure has the lowest stability there and of course it is easy to mount our shaker there. For the response points we want to cover as much of our beam as possible, especially in the vertical beam where the structure will vibrate the most. We have chosen to measure in 9 points distributed along the vertical beam in 3 rounds and 3 points in the horizontal beam, just to get some information of the behavior of that part to have something to compare with the vertical part. Which gives the following model of the geometry, note that 1 represents both the shaker and one response point.

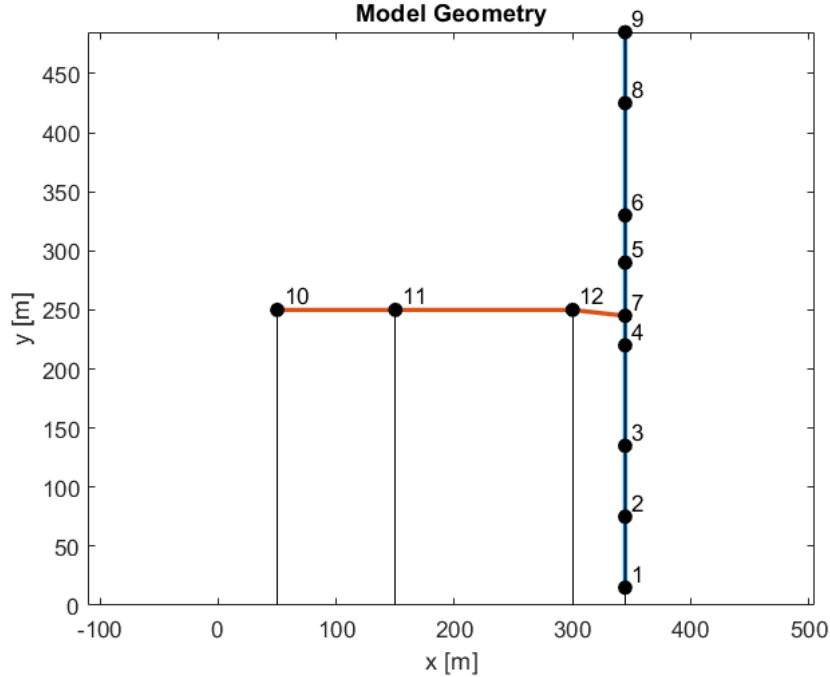


Figure 5: Model geometry

The FE simulation provided mode shapes in 121.55 and 136.18 Hz, to be on the safer side we choose a frequency range from 0-150 Hz in the measurement due to variation of the resonance frequency depending on how precise our measurements is performed.

3 Measurement

The testing setup is the following:



Figure 6: Test setup

The equipment of the lab was the following:

Shaker
3 accelerometers
cabel to accelerometer
Data Acquisition (DAQ)
Computer
QuickDAQ software
Fastening equipment
Wax

Table 2: Lab equipment

The measurement parameters for the lab was:

Range	-10v to 10v
Coupling	AC
Enginnering unit shaker	N
Enginnering unit accelerometer	m/s^2
Sensivity	mv/ms^{-2}
Sensitivity Accelerometer 30415	$9.831 \text{ mV}/(m/s^2)$
Sensitivity Accelerometer 30425	$10.12 \text{ mV}/(m/s^2)$
Sensitivity Accelerometer 30418	$10.06 \text{ mV}/(m/s^2)$
Number of averages	20
Sample rate	4000 Hz
Max frequency	150
FFT size	8192
Response windows type	Hanning
Reference window type	Same as response

Table 3: Measurement parameters

From the measurements we got the following FRF plot and Coherence plot:

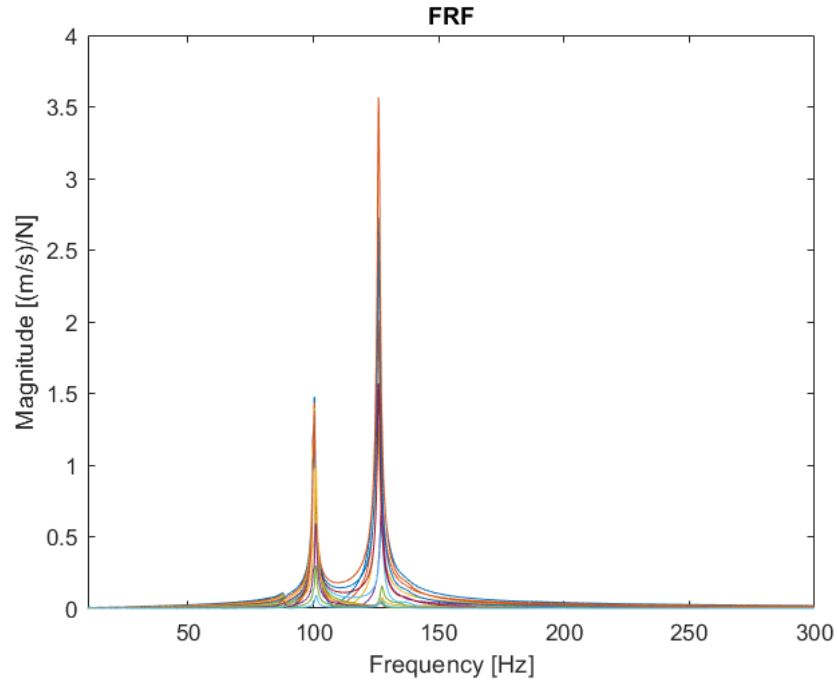


Figure 7: FRF plot of all response points.

The resonance frequencies are approximately 100 and 126 Hz for the measurement data.

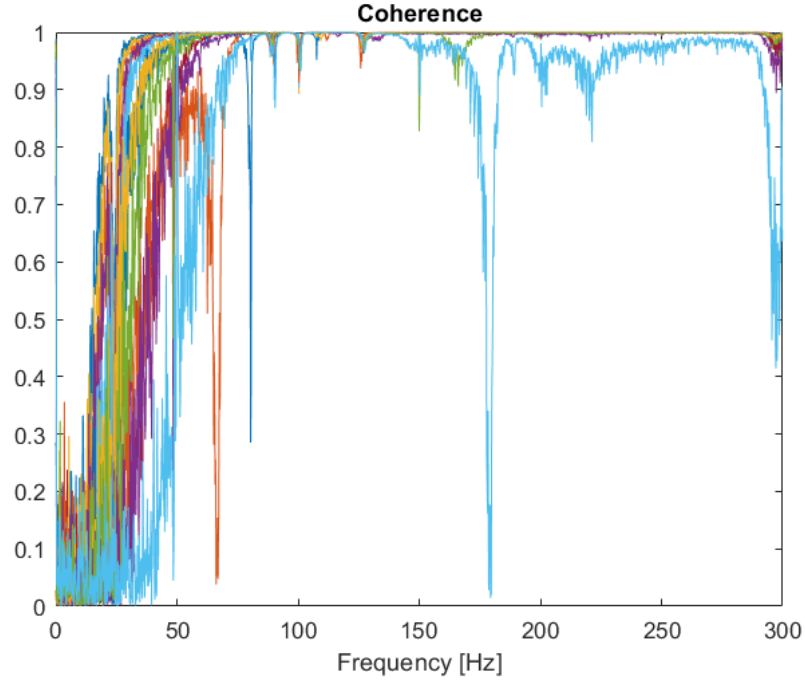


Figure 8: Coherence plot of all response points.

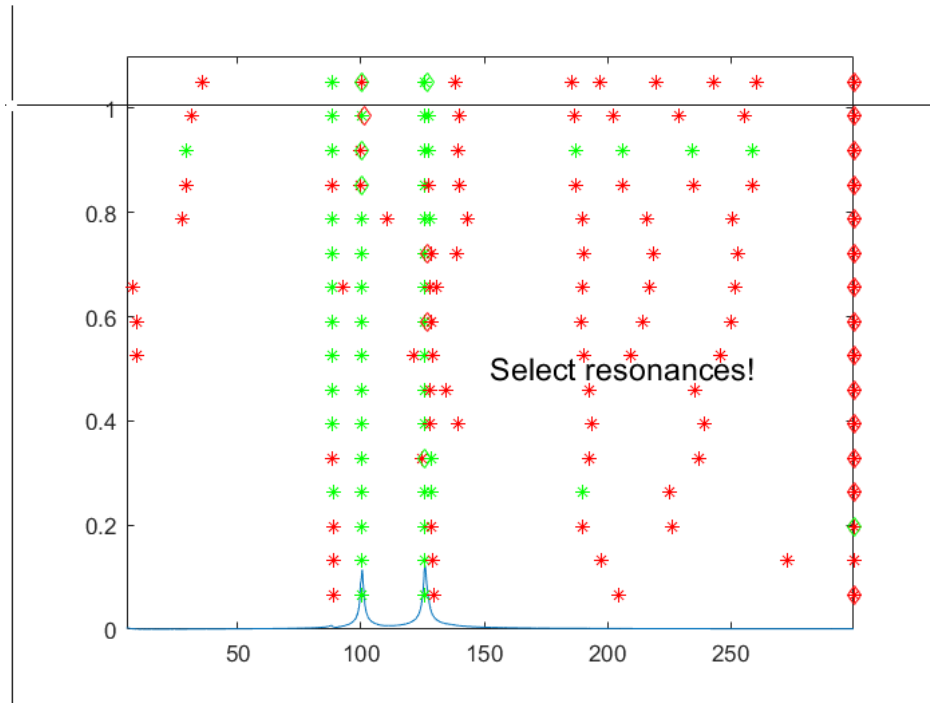


Figure 9: Stabdia Plot

4 Modal analysis and model

From the measurement data, we obtained the first two mode shapes as follows:

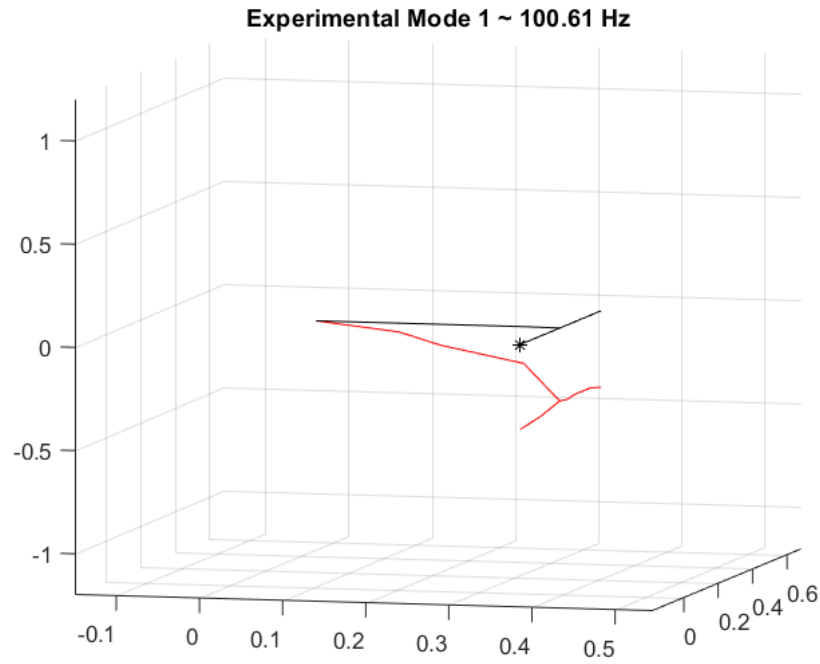


Figure 10: Experimental mode 1.

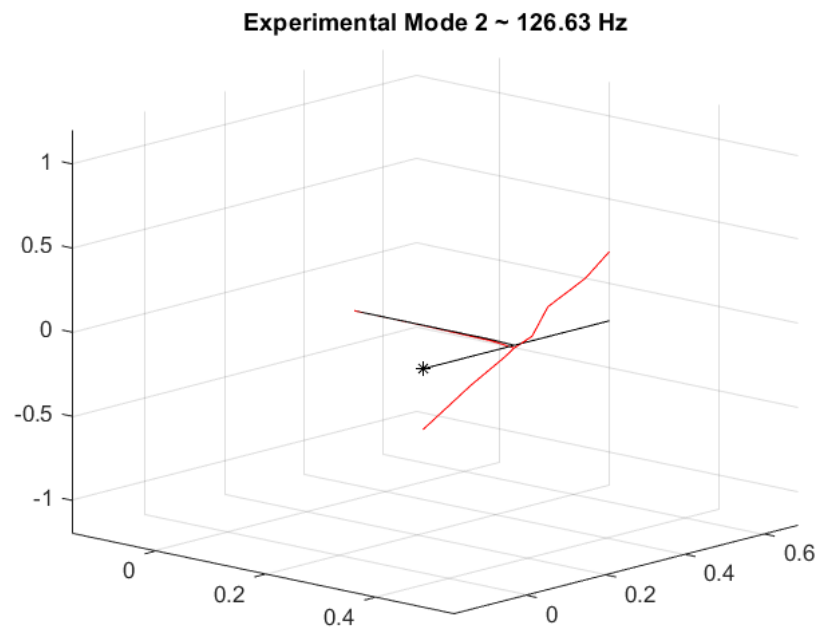


Figure 11: Experimental mode 2.

If we extract the resonance frequencies and the damping values from the experimental data we get the following:

Mode 1: $f = 100.6$ Hz, $\text{zeta} = 0.429$ %
Mode 2: $f = 126.6$ Hz, $\text{zeta} = 0.495$ %

Figure 12: Resonance frequency and damping values for the two first modes.

The resonance frequencies from the modal analysis and the FE-model are:

Mode	FE-model	EMA-model
1	121.55 Hz	100.6 Hz
2	136.18 Hz	125.8 Hz

Table 4: Comparison of resonant frequencies.

5 Updated version

To make the FE-model more accurate we applied springs at the fixed end instead of an ideally fixed support. Since we assumed no motion in x-direction we made it still fixed. Then we applied rotation around x as $1.25 \cdot 10^5$ N/rad, rotation around y as $1.9 \cdot 10^4$ N/rad, rotation around z to 10^6 N/rad and Translation in z as 10^{15} N/m.

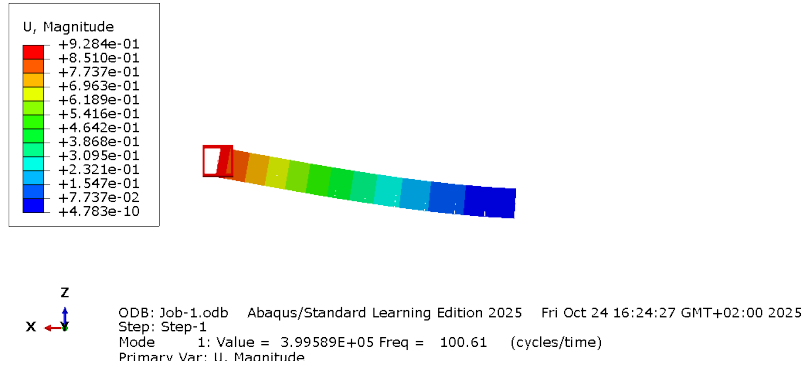


Figure 13: FEA result for Mode 1 after updated clamping conditions.

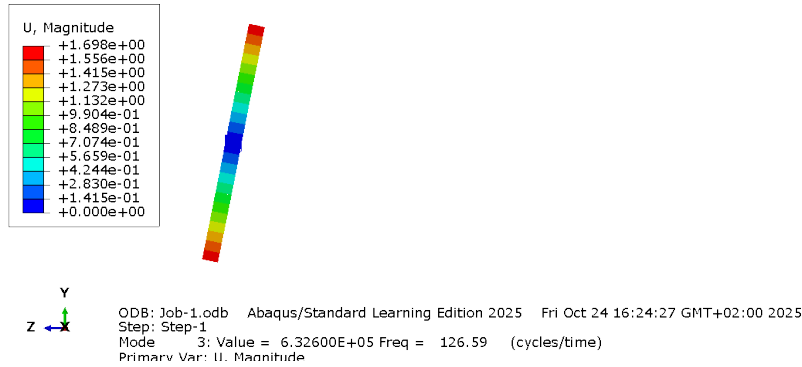


Figure 14: FEA result for Mode 2 after updated clamping conditions.

Mode	First FE-model	Modified FE-model	EMA-model
1	121.55 Hz	100.61 Hz	100.61 Hz
2	136.18 Hz	126.59 Hz	126.63Hz

Table 5: Comparison of resonant frequencies.

The unmodified FE-model gave a slightly higher resonance frequencies. When we applied the springs we nearly got the same resonance frequencies as in the EMA-model. This gives an indications that the EMA-model does not have an ideally fixed end.

Appendix A - Matlab Code

Listing 1: MATLAB script Solving Lab 2 Experimental Data

```

clc; clear; close all;

L = 327; % mm
H = 500; % mm

b = 35;
h = b;

E = 211e9;
rho = 7800;
v = 0.3;

VerticalNodes = [1:9]; % [1, 2, 3, 4, 7, 5, 6, 8, 9]
HorizontalNodes = [10:13];

xVert = (L+b/2)*ones(size(VerticalNodes))*1e-3;
yVert = [15 75 135 220 245 290 330 (H-75) (H-15)]*1e-3; % [15 75 135 220 290 330 (H-255) (H-15)]
zVert = 0*ones(size(VerticalNodes));

xHorz = [300 200 150 50]*1e-3;
yHorz = H/2*ones(size(HorizontalNodes))*1e-3;
zHorz = zeros(size(HorizontalNodes));

x = [xVert xHorz]';
y = [yVert yHorz]';
z = [zVert zHorz]';
nodes = [VerticalNodes HorizontalNodes]';

plotseq = [VerticalNodes 0 5 HorizontalNodes 0];

geodef = [x y z 0*nodes 0*nodes nodes];

measurements = 1:4;
accelerometers = 1:3;

XYZ = geodef(:,1:3)';
coherences = [];
frequencies = [];
mobilities = [];

```

```

data1 = txt2mat( 'Grupp2_1-2-3.csv ');
data2 = txt2mat( 'Grupp2_4-5-6.csv ');
data3 = txt2mat( 'Grupp2_7-8-9.csv ');
data4 = txt2mat( 'Grupp2_10-11-12.csv ');
data5 = txt2mat( 'Grupp2_13.csv ');
datastruct = {data1 data2 data3 data4 data5};

for measurement = measurements
    A = datastruct{measurement};
    f = A(:,1); w = 2*pi*f; s = 1j*w;
    F = complex(A(:, 2), A(:,3));
    Accelerance1 = complex(A(:,5), A(:,6));
    Accelerance2 = complex(A(:,8), A(:,9));
    Accelerance3 = complex(A(:,11), A(:,12));

    Coherence1 = A(:,4);
    Coherence2 = A(:,7);
    Coherence3 = A(:,10);

    Mobility1 = Accelerance1./s;
    Mobility2 = Accelerance2./s;
    Mobility3 = Accelerance3./s;

    coherences = [coherences Coherence1 Coherence3 Coherence2];
    frequencies = [frequencies f f f];
    mobilities = [mobilities Mobility1 Mobility2 Mobility3];
end

%%
close all;

plot(frequencies(:,1), abs(mobilities(:,1)))

%%
figure;
plot(frequencies,abs(mobilities)); xlim([10 300]);
xlabel('Frequency [Hz] ');
ylabel('Magnitude [(m/s)/N] ');
title('FRF');
figure;
plot(frequencies,coherences);
xlabel('Frequency [Hz] ');
title('Coherence');

%%

node = 1;
disptime = 0;
modeNumbers = 1:2;

keep = f>0;

```

```

H_all = mobilities(keep,:);
f_all = f(keep);

flow = 5;
fhi = f_all(end);
%%
hN = 700;
[h,t,fs] = impresp(H_all, f_all);
mif = modeind3(H_all);
mif = mif/max(mif); % normalize
close all;
plot(f_all, mif)
%%
Norders = 30;
figure('Name', 'Complex')
poles = complex(h, fs, hN, Norders, mif, f_all, flow, fhi);
[residues, ~] = pol2resf(H_all, f_all, poles, flow, fhi, disptime);

%%
[f, zeta] = poles2fd(poles); % [Hz, %]

[f, I] = sort(f);
zeta = zeta(I);
residues = residues(I, :, :);

%%
scale = 1/5;
refpnt = 1;
ang = 0;

close all;
figure('Name', 'Geometry'); axis equal; box on; hold on;
title('Model Geometry');
xlabel('x [m]'); ylabel('y [m]');
plot_geometry(geodef, plotseq);

%%
fr = [100.61 126.63];
for m = modeNumbers
    figure('Name', sprintf('Mode-%d', m));
    fprintf('Mode-%d: f=-%7.1f-Hz, -zeta=-%6.3f-%%\n', m, fr(m), zeta(m));
    modevec = residues(m,:);
    modevec = modevec ./ max(abs(modevec));
    modevec = modevec - modevec(13);

    modeplot(gcf, geodef, plotseq, modevec, scale, refpnt, ang);
    title(sprintf('Experimental Mode-%d-~%.2f-Hz', m, fr(m)));
    grid on; axis on; axis normal;

    % [gpoint,G,ref] = animcalc(geodef, plotseq, imag(modevec), 1/10, ...
    %     refpnt);
    % figure('Name', sprintf('Mode %d (animation)', m));
    % animate(gcf, gpoint, G, ref, 5);

```

end

%%

```
function plot_geometry(geodef, plotseq)
    geodef = abs(geodef); plotseq = abs(plotseq);
    lw = 2;

    XY = geodef(:,1:3);
    seq = [];
    for v = plotseq(:).'  
        if v==0  
            if length(seq)>=2  
                plot(XY(seq,1), XY(seq,2), '-', 'LineWidth', lw);  
            end  
            seq = [];  
        else  
            seq = [seq, v];  
        end  
    end

    stem(XY(:,1), XY(:,2), 'ok', 'filled');

    for i=1:size(XY,1)  
        text(XY(i,1), XY(i,2), sprintf('%d', i), 'VerticalAlignment', ...  
            'bottom');  
    end
end
```